Andy's LineMatcher Matching Stack

> The matcher links cities between `shipment report` and `rate card`, scores likely candidates, and decides whether a pair should be auto-matched, sent to review, or left unmatched.

Minimum input requirements

The input workbook should follow this basic contract:

- file format: `.xlsx`
- worksheets named:

- `rc`
- `shrep`

- worksheet `rc` must include:

- `Destination City`
- `Destination Country`

- worksheet `shrep` must include:

- `Delivery City`
- `Arrival Country`

Additional columns are allowed. They are not required for matching itself, but they are preserved in the output sheet `matched_rows`.

The safest input layout is:

- headers in the first row
- one coherent data table per worksheet
- no extra title rows above the table
- exact header names for the required columns

It is also worth noting:

- city and country values do not need to be perfectly standardized, because the matcher normalizes them
- empty values are allowed, but they reduce the chance of a successful match
- if a worksheet name or a required header name changes, the program will not discover that data automatically

Overview

The matcher is deterministic. It does not use ML, embeddings, LLM reasoning, or a trained model.

The stack is:

1. normalize input strings
2. generate RC city variants
3. filter candidates by country
4. check manual alias overrides
5. score each candidate with three fuzzy heuristics
6. keep the best candidate per RC row
7. apply thresholds and confidence margin rules

Matching pipeline

For a single `shrep` row:

1. `Arrival Country` is normalized.
2. `Delivery City` is normalized.
3. The RC candidate pool is restricted to rows with the same normalized `Destination Country`.
4. If `city_aliases.csv` contains an approved override, that mapping wins immediately.
5. Otherwise each candidate gets a fuzzy score.

6. The best three candidates are retained for output and review.
7. The top score and its margin over the second candidate determine the final status.

Normalization layer

`normalize_basic`

Applied first to city names and country values:

- trims whitespace
- converts to uppercase
- removes diacritics with Unicode `NFKD` normalization
- strips non-alphanumeric characters to spaces
- collapses repeated whitespace

Example:

- `Dubai, UAE` -> `DUBAI UAE`
- `Petah-Tikva` -> `PETAH TIKVA`

`normalize_city`

Builds on `normalize_basic` and then:

- removes generic location words such as `CITY`, `TOWN`, `DISTRICT`, `PROVINCE`
- removes a trailing country token when present and when the city still has more than one token

Example:

- `Dubai, UAE` with country `AE` -> `DUBAI`
- `Makati City` -> `MAKATI`

`city_variants`

RC cells may contain multiple location fragments such as:

- `BERKELEY VALE; WARNERVALE NSW`

The matcher splits RC cities on:

- `;`
- `,`

- `/`
- `|`

Each part is normalized and stored as an additional searchable variant that still points to the same original RC row.

Candidate filtering

Before fuzzy scoring, the matcher narrows the pool to:

- candidates where normalized `Arrival Country == Destination Country`

This is a strong safety filter and prevents many wrong matches.

If there are no same-country candidates, the code can optionally fall back to cross-country search, but that behavior is disabled by default.

Alias override

`city_aliases.csv` is a manual memory layer.

If an approved pair exists for:

- normalized source country
- normalized source city

then the matcher returns:

- `status = auto_matched`
- `method = alias`
- `score = 100`

without running fuzzy ranking.

Fuzzy scoring functions

The matcher computes three scores and takes the maximum.

1. Gestalt similarity

Function:

- `sequence_score(left, right)`

Implementation:

- Python `difflib.SequenceMatcher`

Formula:

```
score_seq = 100 * SequenceMatcher(None, left, right).ratio()
```

This is Ratcliff-Obershelp style similarity. It rewards large shared character blocks and works well for spelling variants such as:

- `SHARJACH` vs `SHARJAH`
- `PETAH TIKVA` vs `PETAKH TIKVA`

2. Token sort similarity

Function:

- `token_sort_score(left, right)`

Formula:

- tokenize both strings on spaces
- sort tokens alphabetically
- join back into strings
- apply `sequence_score`

So:

- `PETAH TIKVA`
- `TIKVA PETAH`

become the same sorted token order before the Gestalt score is computed.

This helps when both strings contain the same words but in different order.

3. Subset / containment heuristic

Function:

- `subset_score(left, right)`

Logic:

- convert both strings to token sets
- identify the shorter and longer set
- if the shorter set is a subset of the longer set, assign a high score

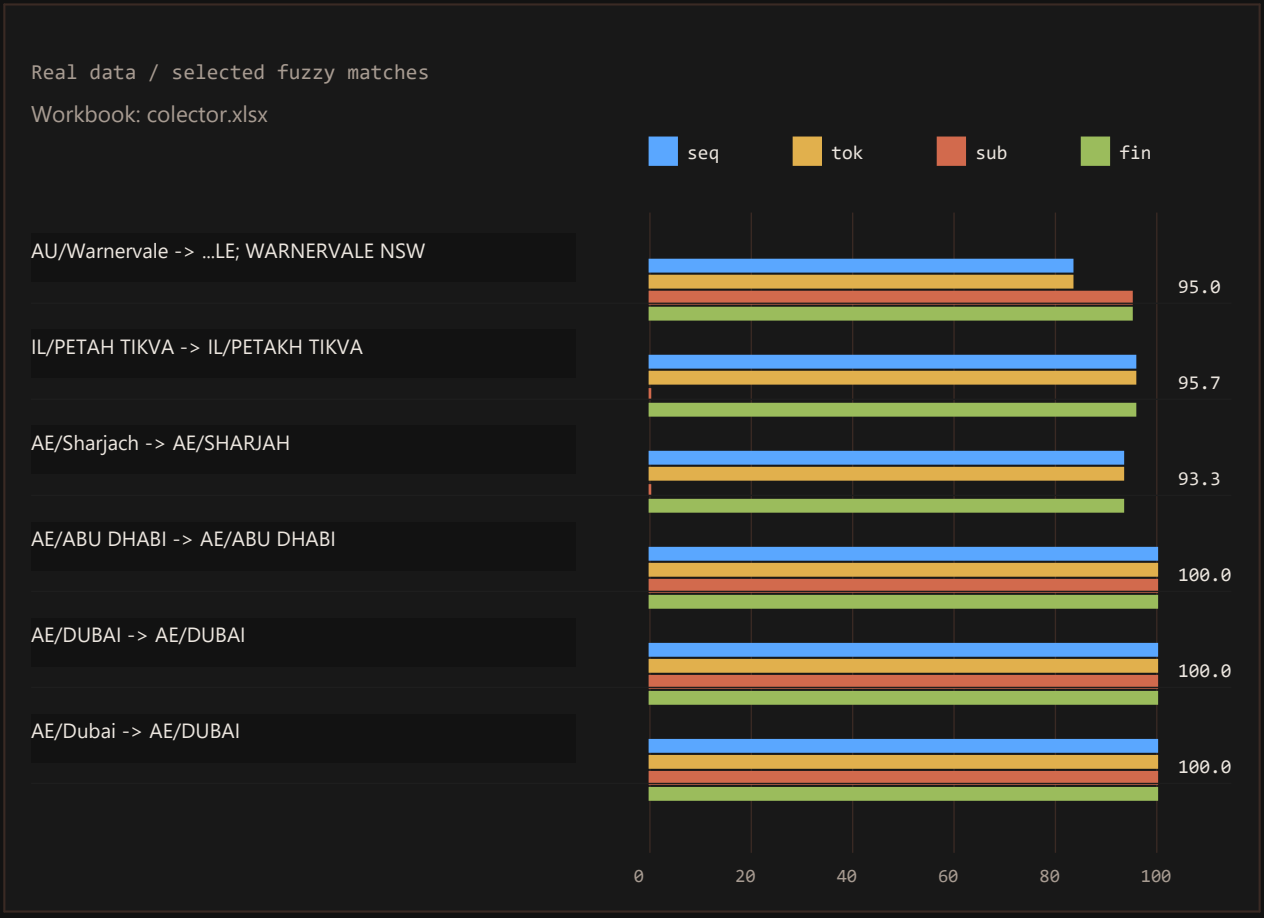
Formula:

- if `shorter` is a subset of `longer`
- `token_gap = len(longer) - len(shorter)`
- `score_subset = max(90, 97 - 2 * token_gap)`
- otherwise `score_subset = 0`

This captures cases where one city is essentially contained inside a longer RC form:

- `WARNERVALE`
- `BERKELEY VALE WARNERVALE NSW`

Final fuzzy score



Function:

- `city_score(left, right)`

Formula:

```
score_final = max(
  score_seq,
  score_token_sort,
  score_subset
)
```

Ranking and de-duplication

Multiple RC variants can point to the same RC row.

To avoid the same RC row appearing several times in the ranking:

- candidates are grouped by `rc_row_id`
- only the highest score for that RC row is kept

Then rows are sorted descending by score.

Decision rules

Real data / top score vs second score

Workbook: colector.xlsx

After ranking:

- `top_score` = score of best candidate
- `second_score` = score of second candidate, or `0`
- `margin = top_score - second_score`

Decision logic:

```
if top_score >= auto_threshold and (top_score == 100 or margin >= min_margin):
    status = auto_matched
elif top_score >= review_threshold:
    status = review_needed
else:
    status = unmatched
```

Default thresholds:

- `auto_threshold = 90`
- `review_threshold = 75`

- `min_margin = 3`

Why this works well for logistics city data

This stack is tuned for:

- spelling differences
- punctuation differences
- country suffixes
- multi-part RC city cells
- variant ordering
- safe human review when certainty is low

It is intentionally auditable. Every accepted match can be explained through:

- normalized source city
- candidate list
- chosen method
- score
- margin

What it is not

It is not:

- Levenshtein distance
- phonetic matching
- semantic matching
- embedding similarity
- machine learning

It is a deterministic fuzzy matcher with manual memory and thresholded review.

Glossary

Fuzzy matching

Text matching that does not require exact character-for-character equality.

It tries to answer:

- do these two strings look similar enough to be treated as the same place

Exact match

Strict matching where both values must be identical after comparison.

Heuristic

A practical scoring rule or simplified decision rule that works well on real-world data.

Deterministic

Given the same input data and the same thresholds, the program always returns the same result.

Normalization

Standardizing text before comparison, for example by:

- uppercasing
- stripping punctuation
- collapsing whitespace
- removing diacritics

Token

A single text unit, usually a word after splitting a normalized string.

Tokenization

Splitting text into tokens.

Gestalt similarity

Character-block similarity based on large shared sequence fragments.

In this project it is implemented with:

- `difflib.SequenceMatcher`

Ratcliff-Obershelp

A family of sequence similarity ideas centered on finding the largest shared chunks between two strings.

Token sort similarity

A method that:

1. splits both strings into tokens
2. sorts tokens alphabetically
3. compares the sorted strings

This helps when the same words appear in a different order.

Subset / containment

A heuristic that checks whether the shorter token set is fully contained in the longer one.

Candidate ranking

A sorted list of possible RC matches, ordered from strongest to weakest.

Candidate

A single possible RC record that may match the source city.

`top_score`

The best score in the candidate ranking.

`second_score`

The second-best score in the candidate ranking.

Margin

The difference between the best and second-best score:

- `margin = top_score - second_score`

Threshold

A decision cutoff used to route a result into auto-match, review, or unmatched.

Auto match

A match strong enough to be accepted automatically.

Manual review

A case where the matcher sees a plausible candidate, but not with enough confidence to auto-accept.

Alias override

A manual mapping stored in `city_aliases.csv` that bypasses fuzzy scoring.

Embedding similarity

Similarity computed from vector representations of text rather than only from raw characters.

In practice:

- a model turns text into numeric vectors
- similarity is computed between those vectors

This can capture semantic closeness, but it is heavier and less transparent than the current heuristic stack.

Semantic similarity

Similarity based more on meaning than on literal spelling.

Exonym

A place name used in another language that differs from the local name.

Examples:

- `Munich` for `München`
- `Vienna` for `Wien`

Endonym

The local native name of a place.

Transliteration

Writing the same name from another script using Latin letters.

Diacritics

Characters such as:

- `é`
- `ö`
- `ą`

These are often removed during normalization.

Cross-country fallback

An optional fallback path where the matcher could search outside the same country when no same-country candidate exists.

It is disabled by default in this project because it increases false-match risk.